

Kunal Maheshwari

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Summary

Java Backend / AI-Enabled Software Engineer with 4+ years delivering high-availability healthcare and automotive systems, plus hands-on project use of Codex and GitHub Copilot to build, review, debug, test, and document software.

Delivery scope: Translates requirements and design constraints into structured implementation plans, Java/API and database changes, tests, review evidence, CI/CD readiness, release validation, and production support.

Engineering approach: Treats AI output as untrusted, reviews it for correctness and maintainability, applies relevant security constraints, and accepts changes only after builds, automated tests, execution checks, and runtime validation.

Working style: Takes ownership with transparent risk communication, collaborates across architecture, QA, database, deployment, and operations teams, and improves engineering consistency through reusable instructions, documentation, code reviews, and mentoring.

Technical Skills

Java/Computer Science: Java 8/11, Core Java, OOP, Collections, Streams, design patterns, data structures and algorithms, system and component design, REST/SOAP APIs, Spring Framework/MVC, Java EE/J2EE, Hibernate/HQL, JDBC/OJDBC; networking and concurrency (knowledge).

AI-Enabled Engineering (projects): Codex, GitHub Copilot, prompt engineering, custom instruction authoring, specification-driven agent workflows, human-in-the-loop code review, AI-assisted debugging/testing/documentation, LangChain, Ollama, RAG, Qdrant.

Architecture/Cloud: Enterprise SOA and distributed-service integration, secure coding and dependency modernization, API integration, messaging, performance tuning, observability, failure handling; Azure Service Bus/Relay/Functions (professional), AWS and Docker/Terraform (project).

Data/Delivery: Oracle, DB2, PostgreSQL (project/training); Jenkins, Maven, IBM UrbanCode Deploy, GitLab CI/CD, Git, GitHub Actions; JUnit/JUnit4, integration testing, SoapUI, Postman, Log4j2, Agile/Scrum.

Work Experience

NTT Data, Bengaluru, India

Software Developer, Aug 2022 - Aug 2024

- Developed and maintained **Java 8/11, Spring Framework/JAX-WS, and REST/SOAP** healthcare payer services on IBM WebSphere, supporting **50K+ daily requests at 99.99% SLA availability**.
- Integrated services across Azure Relay, Azure Service Bus, Azure Functions, and JMS, plus synchronous downstream REST dependencies with explicit timeout and partial-failure behavior.
- Integrated Java services with Oracle through OJDBC, consuming established views and PL/SQL procedures while applying member/provider rules and validating downstream data propagation.
- **Led a project-wide Log4j2 migration** prompted by security review, owning the PoC, dependency and configuration decisions, preservation of logging contracts, reusable guidance, code reviews, and coordinated rollout.
- Integrated AOP-based service monitoring that propagated correlation IDs and captured **latency, success/failure, and request-volume telemetry** for operational diagnosis.
- Supported root-cause analysis of a WebSphere memory/resource-retention incident through code-path review, mitigation strategy, and rolling-restart validation across four load-balanced production nodes.
- Coordinated requirements, implementation, JUnit/SoapUI testing, reviews, Jenkins/Maven/UrbanCode workflows, release checks, documentation, and go/no-go or rollback recommendations.

Capgemini, Mumbai, India

Senior Analyst, Jul 2020 - Aug 2022

- Developed and supported a Java EE/J2EE, JSF, Hibernate/HQL application on Oracle WebLogic with DB2 for Mercedes/Daimler Vans configuration, recommendation, and ordering workflows.

- Integrated a Python prediction service through asynchronous Java REST requests and callbacks, retaining deterministic option/package validation in Java before recommendations reached dealer users.
- **Reduced VIN repeat-configuration API latency from 60+ seconds to 10–20 seconds** through set-based HQL/DB2 mapping while retaining authoritative business rules in Java.
- Diagnosed defects across Python output, Java validation logs, API behavior, and DB2 reference data; used Postman and production logs to isolate failures and verify performance.
- Supported Jenkins/GitLab CI/CD, Git code reviews, JUnit4 testing, Agile delivery, and documentation; mentored junior developers through walkthroughs, debugging, and reviews.

Capgemini, Mumbai, India

Analyst Trainee, Jan 2020 - Jun 2020

- Built Java Spring/Spring MVC, Angular, and PostgreSQL training applications with authentication, email/OTP login, CRUD, balances, transactions, and profile workflows.

Selected Projects

Resume Corpus - Specification-Driven AI Engineering Workflow

- Directed AI-assisted implementation with **Codex and GitHub Copilot**, authoring evidence constraints, structured agent instructions, reusable workflows, and acceptance criteria for an isolated resume optimization system.
- Built deterministic validators, explainable scorecards, regression cases, build/parse checks, and human acceptance gates; reviewed generated changes and rejected outputs that violated evidence or formatting rules.

CodingHelper - Agentic Java/Python Learning Platform

- Built a React/TypeScript and FastAPI platform using **LangChain, Ollama/Qwen 2.5 Coder, RAG, Qdrant, and PostgreSQL** to generate and retrieve structured Java/Python DSA solution ladders.
- Implemented isolated code execution and repair, rejection, and promotion gates; AI-generated implementations must pass bounded asserting tests before display or reuse.

AWS Automated Spark Pipeline

GitHub

- Provisioned an event-driven S3, Lambda, EFS, ECS, RDS PostgreSQL, VPC/IAM, Docker, Terraform, and CloudWatch pipeline that processed an 8GB+ dataset with persisted checkpoints for restart recovery.

Education

Central Michigan University, Mount Pleasant, MI, USA

Aug 2024 - Dec 2025

Master of Science in Computer Science, GPA: 3.87/4.0

Lovely Professional University, Punjab, India

Aug 2016 - May 2020

Bachelor of Technology in Computer Science and Engineering, CGPA: 8.02/10

Certifications

Microsoft Certified: Azure Fundamentals | Azure Developer Associate | Azure Data Engineer Associate